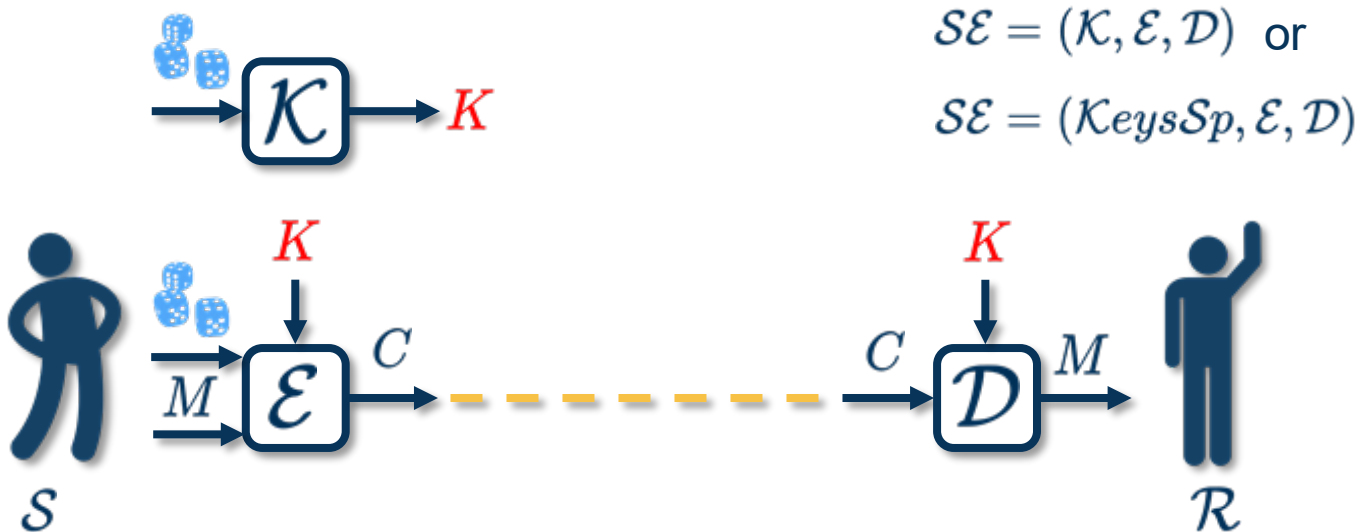


# **SYMMETRIC ENCRYPTION**

## **ENCRYPTION MODES**

## **SECURITY NOTIONS**

- ◆ A scheme  $\mathcal{SE}$  is specified by 3 algorithms  $\mathcal{K}$  (or  $\mathcal{KeySp}$ ),  $\mathcal{E}$ ,  $\mathcal{D}$  and the message space  $\mathcal{MsgSp}$ .



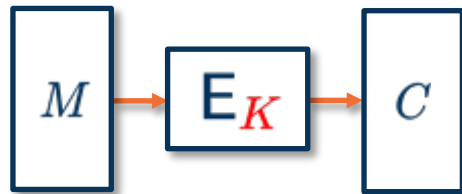
It is required that for every  $M \in \mathcal{MsgSp}$  and every  $K \in \mathcal{KeySp}$  (or output by  $\mathcal{K}$ ),  $\mathcal{D}(K, \mathcal{E}(K, M)) = M$ .

- ◆ Often the key generation algorithm simply picks a random string from some key space  $\mathcal{KeySp}$  (e.g.  $\{0, 1\}^k$  for some integer  $k$  ) .
  - In this case we will say that a scheme  $\mathcal{SE}$  is defined by  $\mathcal{KeySp}$  and two algorithms:  $\mathcal{SE} = (\mathcal{KeySp}, \mathcal{E}, \mathcal{D})$
- ◆ The encryption algorithm can be
  - randomized (take as input a random string)
  - stateful (take as input some state (e.g. counter) that it can update)

# Blockciphers, the main tool of symmetric crypto

## Examples: DES, 3DES, AES...

- A tool to encrypt short strings
- A block cipher is a function family  $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ , where  $k$  (key length),  $n$  (input and output lengths) are the parameters



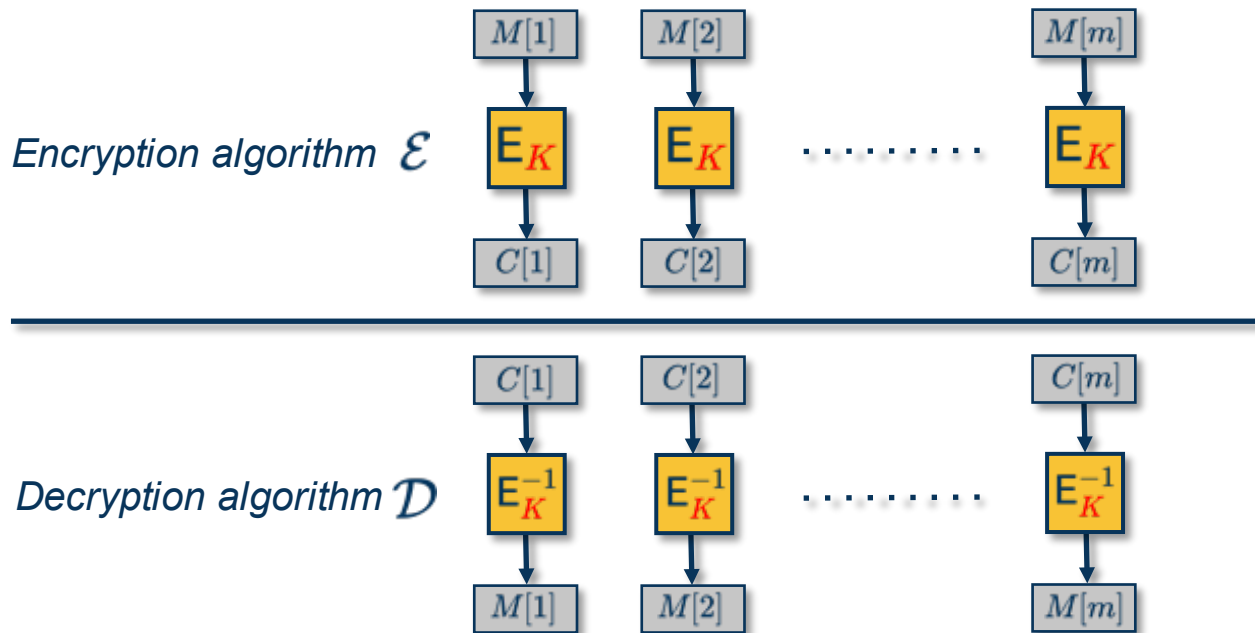
- Notation: for every  $K \in \{0, 1\}^k$ ,  $E_K : E(K, M)$
- For every  $K \in \{0, 1\}^k$ ,  $E_K(\cdot)$  is a permutation (one-to-one and onto function) on  $\{0, 1\}^n$   
For every  $C \in \{0, 1\}^n$  there is a single  $M \in \{0, 1\}^n$  s.t.  $C = E_K(M)$
- Each blockcipher has an inverse denoted  $E_K^{-1}(\cdot)$  s.t.  $E_K(E_K^{-1}(C)) = C, E_K^{-1}(E_K(M)) = M$  for all  $M, C \in \{0, 1\}^n$

# Blockcipher modes of operation

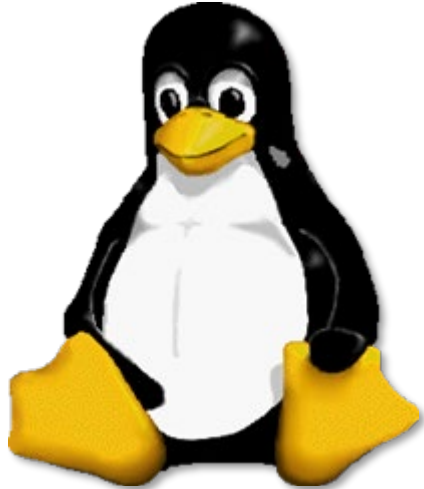
- ◆ Modes of operation define how to use a block cipher to encrypt long messages
- ◆ For simplicity we will often assume that the message space consists of messages whose length is multiple of a block length (otherwise, messages can be padded)

# Electronic Code Book (ECB) mode

- Let  $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$  be a block cipher.  $ECB = (\{0, 1\}^k, \mathcal{E}, \mathcal{D})$ :



# Encrypt an image with ECB



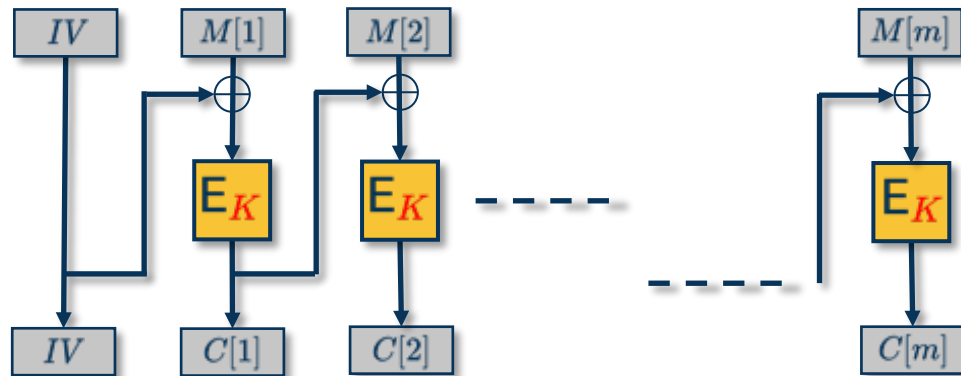
# Cipher-block chaining (CBC) mode with random IV

Let  $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$  be a block cipher.

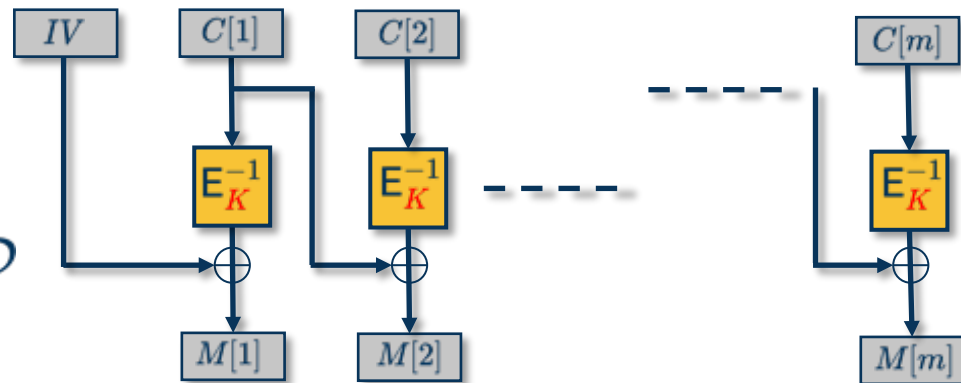
$\text{CBC\$} = (\{0, 1\}^k, \mathcal{E}, \mathcal{D})$ :

*Encryption algorithm  $\mathcal{E}$*

$IV \xleftarrow{\$} \{0, 1\}^n$



*Decryption algorithm  $\mathcal{D}$*



# Stateful cipher-block chaining (CBC) mode with counter IV

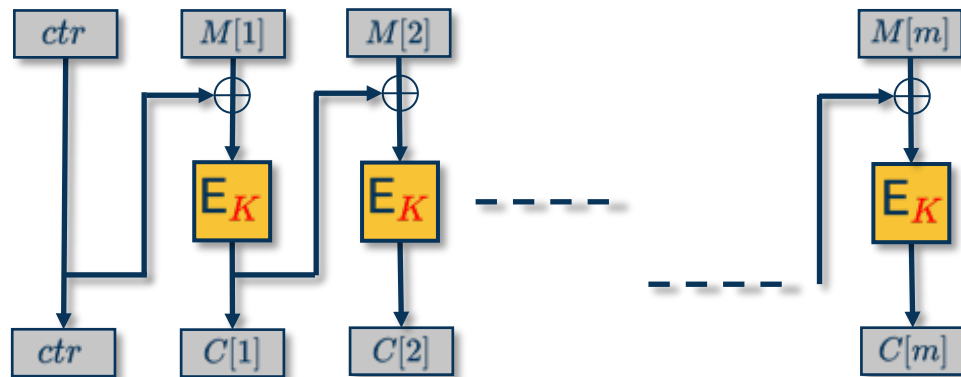
Let  $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$  be a block cipher.

$\text{CBCC} = (\{0, 1\}^k, \mathcal{E}, \mathcal{D})$ :

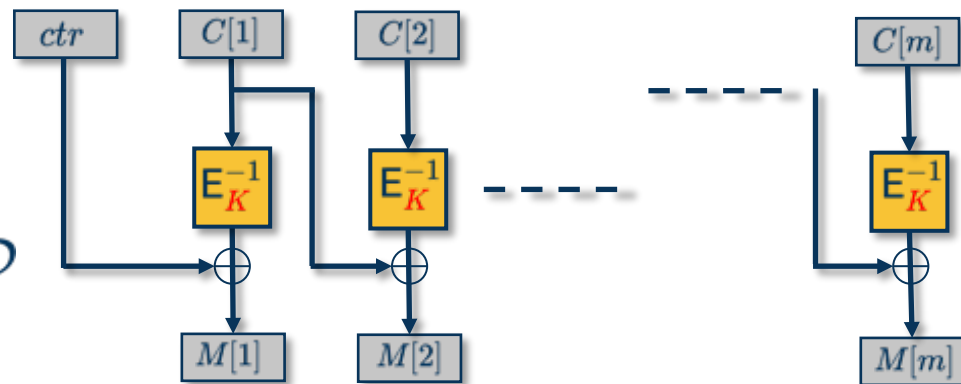
*Encryption algorithm  $\mathcal{E}$*

The counter is incremented for each message until it wraps around

$ctr \leftarrow 0^n$



*Decryption algorithm  $\mathcal{D}$*



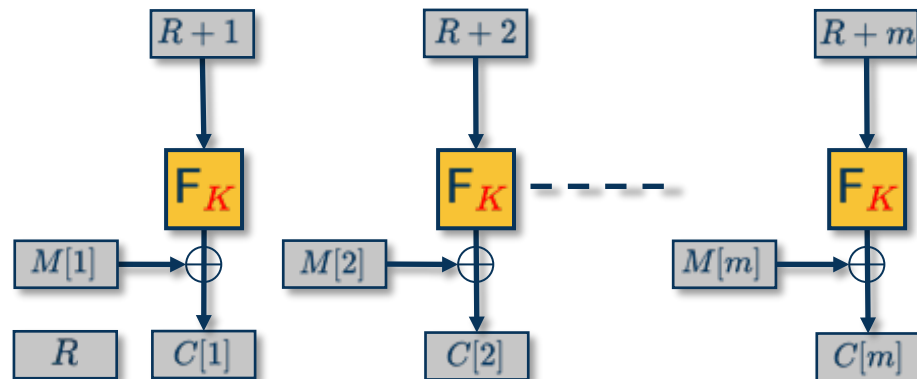
# Randomized counter mode (CTR\$)

Let  $F : \{0, 1\}^k \times \{0, 1\}^l \rightarrow \{0, 1\}^L$  be a function family.

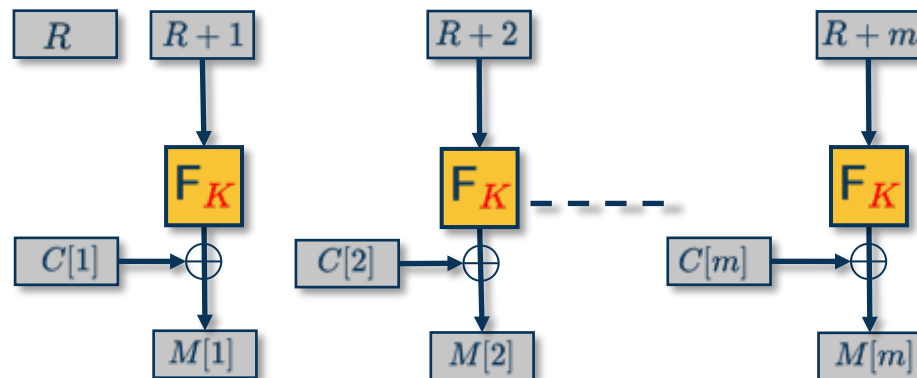
CTR\$ = ( $\{0, 1\}^k$ ,  $\mathcal{E}$ ,  $\mathcal{D}$ ) :

Encryption algorithm  $\mathcal{E}$

$R \xleftarrow{\$} \{0, 1\}^l$



Decryption algorithm  $\mathcal{D}$



# Stateful counter mode (CTR)

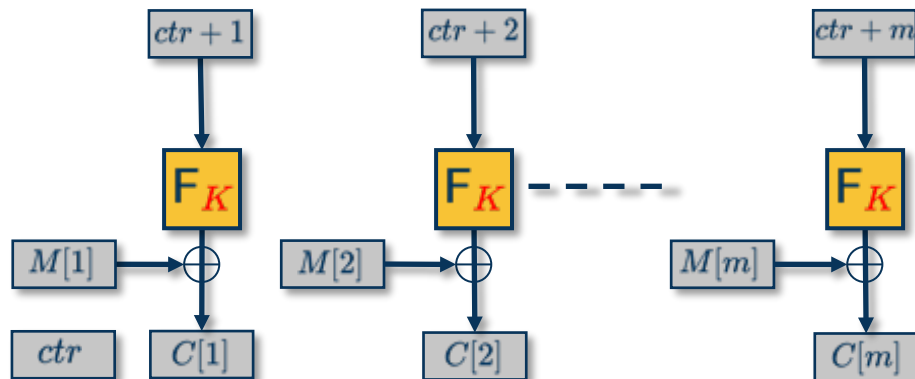
Let  $F : \{0, 1\}^k \times \{0, 1\}^l \rightarrow \{0, 1\}^L$  be a function family.

CTRC =  $(\{0, 1\}^k, \mathcal{E}, \mathcal{D})$  :

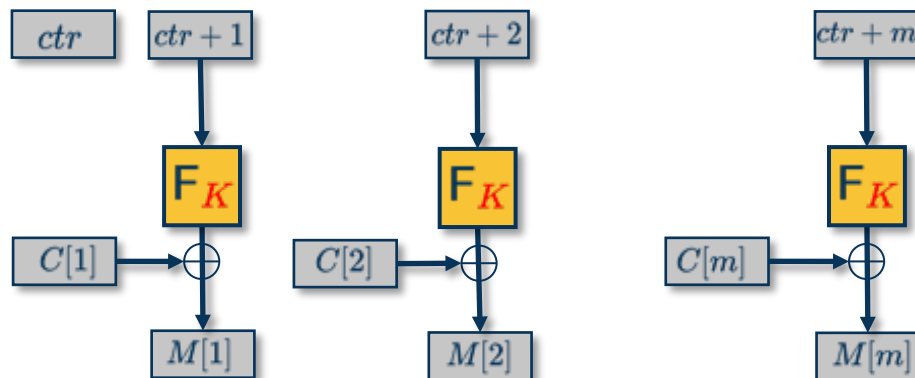
A current counter is  $ctr$  maintained as a state

Encryption algorithm  $\mathcal{E}$

$ctr$  is initially  $0^l$



Decryption algorithm  $\mathcal{D}$



# What is a secure encryption scheme?

- ◆ Recall, perfectly secure schemes are impractical
- ◆ We assume that adversaries are computationally bounded
- ◆ A scheme is secure when it is not insecure
- ◆ Insecure = adversaries can do bad things
- ◆ Bad things: an adversary, who sees ciphertexts
  - can compute the secret key
  - can compute some plaintexts
  - can compute the first bit of a plaintext
  - can compute the sum of the bits of a plaintext
  - can see when equal messages are encrypted
  - can compute ...

# So what is a secure encryption scheme?

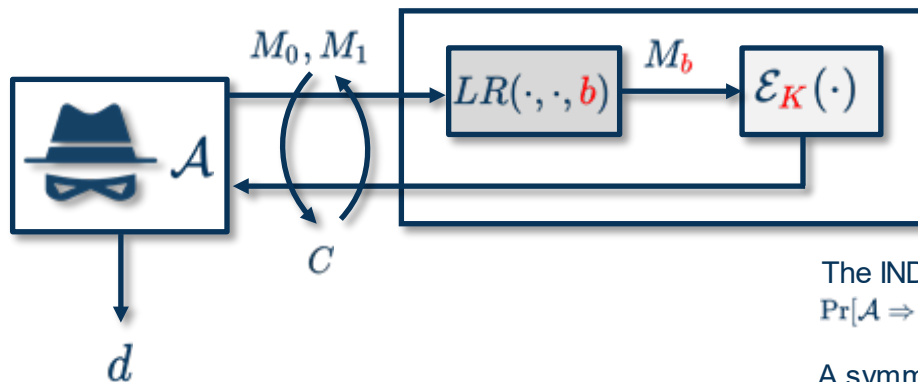
- Informally, an encryption scheme is secure if no adversary with “reasonable” resources who sees several ciphertexts can compute any\* partial information about the plaintexts, besides some a-priori information.  
\* Any information, except the length of the plaintexts. We assume the length of the plaintexts is public.
- Note, that the above implies that the bad things we mentioned do not happen. And the other “bad” things.
- While the above “definition” captures the right intuition, it’s too informal to be useful.

# Indistinguishability under chosen-plaintext attacks

Fix  $\mathcal{SE} = (\text{KeySp}, \mathcal{E}, \mathcal{D}), K \xleftarrow{\$} \text{KeySp}$

For an adversary  $\mathcal{A}$  and a bit  $b$  (either 0 or 1) consider an experiment  $\text{ind-cpa-}b$ , i.e., consider two experiments,  $\text{ind-cpa-0}$  (“left”) and  $\text{ind-cpa-1}$  (“right”).

In each experiment,  $\mathcal{A}$  is given access to the “left-right” encryption oracle that takes two messages and returns encryption of  $M_b$  under  $K$



The IND-CPA advantage of  $\mathcal{A}$  (denoted as  $\text{Adv}^{\text{ind-cpa}}(\mathcal{A})$ ) is  $\Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cpa-0 exp.}] - \Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cpa-1 exp.}]$

A symmetric encryption scheme  $\mathcal{SE}$  is indistinguishable under chosen-plaintext attacks (IND-CPA secure) if for any adversary  $\mathcal{A}$  with “reasonable” resources its ind-cpa advantage is “small” (close to 0).

# Resources of an adversary

1. Time-complexity is measured in some fixed RAM model of computation and includes the maximum of the running-times of  $\mathcal{A}$  in the experiments, plus the size of the code for  $\mathcal{A}$ .
2. The number of queries  $\mathcal{A}$  makes.
3. The total length of all queries.

# How to prove a scheme is not secure

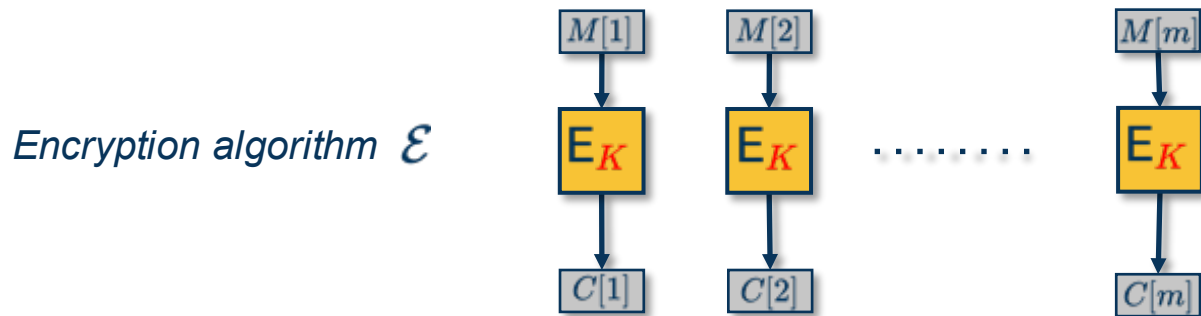
1. Construct an adversary (present a pseudocode for its algorithm) that breaks the scheme(e.g. ECB) according to the definition of security in question (e.g. IND-CPA).
2. Calculate the adversary's advantage. It shouldn't be too close to 0. E.g. something like  $1/2^{60}$  is too small.  $1/1000$  is OK (for the adversary, not security).
3. Calculate the adversary's resources (time, number of queries, total number of bits in queries). They should be "reasonable". E.g.,  $2^{60}$  queries is not reasonable.

# Let's test IND-CPA definition

- ◆ Consider various toy examples.

# Analysis of the ECB mode

Let  $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$  be a block cipher.  $\text{ECB} = (\{0, 1\}^k, \mathcal{E}, \mathcal{D})$ :



- ◆ Is ECB a good encryption scheme?
- ◆ Is ECB IND-CPA secure?

- ◆ **Claim**: any deterministic, stateless scheme is not IND-CPA.

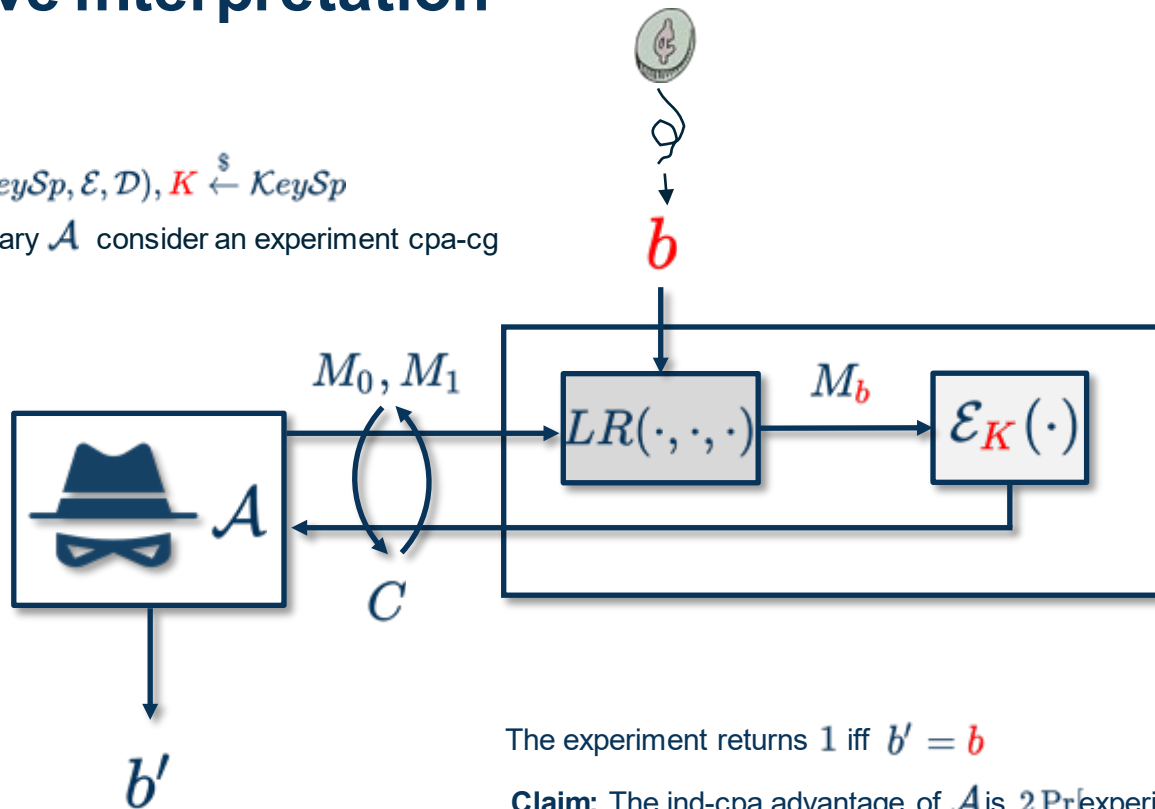
- ◆ Why?

- ◆ Turns out that the rest of the modes are IND-CPA secure assuming that the underlying blockcipher is good.
- ◆ Before that we will learn what does it mean for a blockcipher to be good.

# Alternative interpretation

Fix  $\mathcal{SE} = (\text{KeySp}, \mathcal{E}, \mathcal{D}), K \xleftarrow{\$} \text{KeySp}$

For an adversary  $\mathcal{A}$  consider an experiment cpa-cg



The experiment returns 1 iff  $b' = b$

Claim: The ind-cpa advantage of  $\mathcal{A}$  is  $2 \Pr[\text{experiment cpa-cg returns 1}] - 1$